LIFTING AGAINST INJECTIVE RING MAPS CHARACTERIZES CLOSED IMMERSIONS FOR QUASI-COMPACT MORPHISMS

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Editorial note. Counted span is the author's own proof environment, source lines 23612-23658, immediately following the statement at 23598-23610.

Editorial note. The author writes 'details omitted' at source line 23654 for the retraction step; that sentence is retained verbatim as required, and the surrounding argument (freely used epimorphism and the split-mono affinity criterion) is fully displayed in the span.

Editorial note. Cites lemma-affine-extensive-criterion and lemma-scheme-affine-if-aff-is-split-mono by \ref; both are navigation pointers into the same chapter.

Allowed standard prerequisites (editorial).

- Affine form of the extensive criterion. chapter lemma lemma-affine-extensive-criterion, cited at proof lines 23629 and 23656. Supplies the affine case in both directions: the lift over an affine base and the conclusion that the morphism is a closed immersion.
- A scheme is affine when the canonical map to the spectrum of its global sections is a split monomorphism. chapter lemma lemma-scheme-affine-if-aff-is-split-mono, cited at proof line 23655. Converts the retraction h produced by the lifting property into affineness of X.
- Closed immersions are local on the target. standard Stacks fact used at proof line 23633. Reduces the converse direction to an affine base.
- Quasi-compactness yields a finite affine open covering. used at proof line 23636. Produces the finite affine cover U_1, ..., U_n of X and the affine coproduct U used in the lifting square.

Source reference. Original lemma 0H2P: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23496–23496. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0H2Q: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23547–23547. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a quasi-compact morphism of schemes. Then f is a closed immersion if and only if for every injective ring map $A \to B$ and*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

commutative square

$$\begin{array}{ccc} \mathrm{Spec}(B) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

there exists a lift $\mathrm{Spec}\, A \to X$ *making the diagram commute.*

Proof. Assume that f is a closed immersion. Let $A \to B$ be an injective ring map and consider a commutative square

$$\begin{array}{ccc} \mathrm{Spec}(B) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

Then $\mathrm{Spec}(A) \times_Y X \to \mathrm{Spec}(A)$ is a closed immersion and hence we get an ideal $I \subset A$ and a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(B) & \longrightarrow & \mathrm{Spec}(A/I) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow & & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

We obtain a lift by Lemma 0H2P.

Assume that f has the lifting property stated in the lemma. To prove that f is a closed immersion is local on Y, hence we may and do assume Y is affine. In particular, Y is quasi-compact and therefore X is quasi-compact. Hence there exists a finite affine open covering $X = U_1 \cup \ldots \cup U_n$. The source of the morphism

$$\pi : U = \coprod U_i \longrightarrow X$$

is affine and the induced ring map $\Gamma(X, \mathcal{O}_X) \to \Gamma(U, \mathcal{O}_U)$ is injective. By assumption, there exists a lift in the diagram

$$\begin{array}{ccc} U & \xrightarrow{\pi} & X \\ \downarrow & \overset{h}{\nearrow} & \downarrow f \\ \mathrm{Spec}(\Gamma(X, \mathcal{O}_X)) & \xrightarrow{f'} & Y \end{array}$$

where f' is the morphism of affine schemes corresponding to the ring map $\Gamma(Y, \mathcal{O}_Y) \to \Gamma(X, \mathcal{O}_X)$. It follows from the fact that π is an epimorphism that the morphism h is a retraction of the canonical morphism $X \to \mathrm{Spec}(\Gamma(X, \mathcal{O}_X))$; details omitted. Hence X is affine by Lemma 0H2Q. By Lemma 0H2P we conclude that f is a closed immersion. □